

React Application — Production Deployment

DevOps Project Submission Document | Author: Sudirpandian | Date: June 2026

GitHub Repository: <https://github.com/sudirpandian/Devops/tree/project-2> | **Branch:** project-2

1. Project Overview

This document covers the end-to-end production deployment of the Trend React application, including containerization, infrastructure provisioning, Kubernetes deployment, CI/CD pipeline, and monitoring.

Component	Technology	Details
Application	React (Vite build)	Served via Nginx on port 3000
Container	Docker	sudirpandian01/trend-app
Infrastructure	Terraform	AWS VPC, EC2, Security Group, IAM
Kubernetes	AWS EKS	trend-eks, ap-south-1, 2 nodes
CI/CD	Jenkins	Declarative pipeline with GitHub webhook
Monitoring	Prometheus + Grafana	kube-prometheus-stack via Helm

2. Application Setup

2.1 Clone Repository

```
git clone https://github.com/Vennilavanguvi/Trend.git
cd Trend
ls -la
```

```
[root@ip-172-31-11-132 project]# tree Trend/
```

```
Trend/
├── README.md
└── dist
    ├── assets
    │   ├── About-DbHf8GFx.js
    │   ├── Cart-CKKAW61b.js
    │   ├── CartTotal-4jDB2azO.js
    │   ├── Collection-CSvJy8r8.js
    │   ├── Contact-BomtQVpG.js
    │   ├── ForgotPassword-B8rsrKs1.js
    │   ├── Home-Bjlx AQU1.js
    │   ├── Login-B3PL5z7p.js
    │   ├── NewsletterBox-h8A4Pb2y.js
    │   ├── Orders-CaCJbx18.js
    │   ├── PlaceOrder-D9bDxOFB.js
    │   ├── Product-gEYDQbwF.js
    │   ├── ProductItem-CABqN4-8.js
    │   ├── Profile-C4iJY7-d.js
    │   ├── ResetPassword-Dp6S-2CI.js
    │   ├── Title-D4NmpLxE.js
    │   ├── WishList-D8fx7B72.js
    │   ├── about_img-BAJyTXw9.png
    │   ├── contact_img-CyOum2vk.png
    │   ├── hero_img-D0COb6wn.png
    │   ├── index-B52p61-W.css
    │   ├── index-BjbOuBLy.js
    │   ├── logo-BDGV9g83.png
    │   ├── p_img1-BTuXsnJ1.png
    │   ├── p_img10-Bw8q3bg_.png
    │   ├── p_img11-BSK5z8O8.png
    │   ├── p_img12-CWT58dgm.png
    │   ├── p_img13-9keE2fyF.png
    │   ├── p_img14-3-oykR1l.png
    │   ├── p_img15-CEqmdW8L.png
    │   ├── p_img16-DpIAWCPe.png
    │   ├── p_img17-D3TjESyL.png
    │   ├── p_img18-C8UsXTNh.png
    │   ├── p_img19-Ddd8N1Sm.png
    │   ├── p_img20-BN74rvum.png
    │   ├── p_img21-BhDxXaVx.png
    │   ├── p_img22-DIBExxK1.png
    │   ├── p_img23-CGTC-19z.png
    │   ├── p_img24-DxfyHthA.png
    │   ├── p_img25-DpcquDQp.png
    │   ├── p_img26-DAHi2sy3.png
    │   ├── p_img28-CJDN8kul.png
    │   └── p_img2_1-ku4Fyxmj.png
    └──
```

Emoji

Undo

Cut

Copy

Paste

Paste as plain text

Select all

Writing direction

Visual Search

More tools

Inspect

```
├── p_img48-DwPRHR8.png
├── p_img49-CghJxhG1.png
├── p_img5-BMHJXi1.png
├── p_img50-Cg-7CHi5.png
├── p_img51-FROODGKA.png
├── p_img52-CY-YbGLu.png
├── p_img6-B2VXR-Iw.png
├── p_img7-Ds_MlPCY.png
├── p_img8-BeUACWw0.png
├── p_img9-DmR0Ahy1.png
├── razorpay_logo-DrY6y6Wi.png
├── index.html
└── vite.svg
```

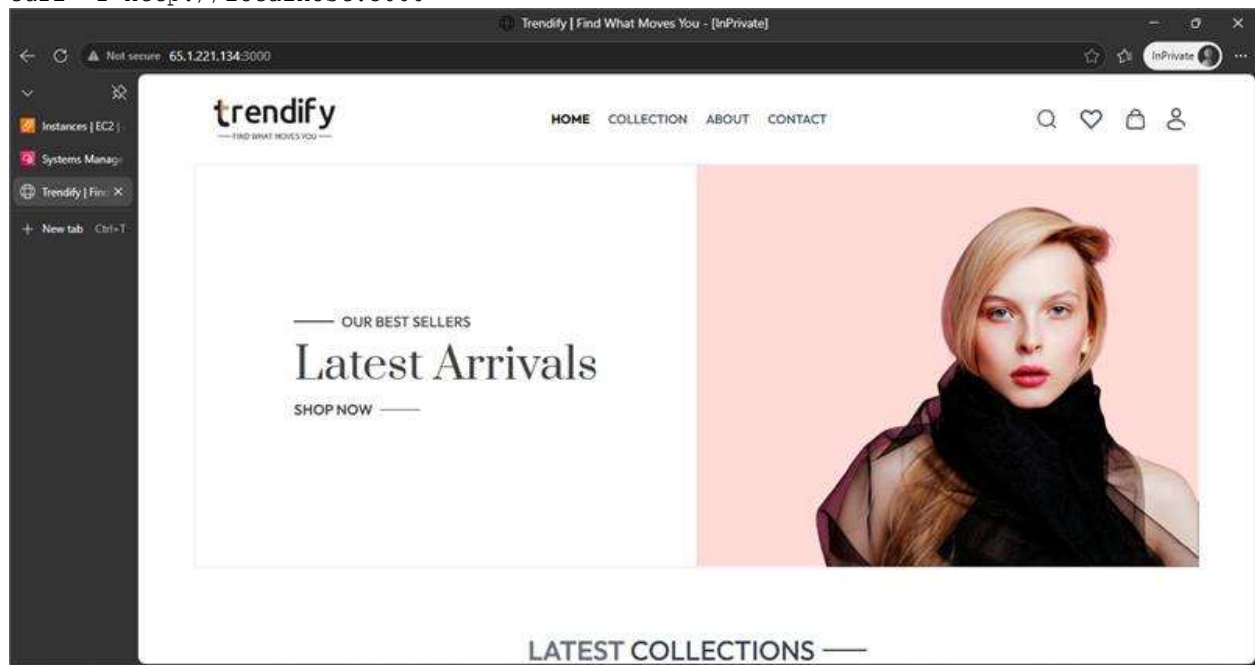
```
directories, 74 files
```

```
[root@ip-172-31-11-132 project]# cd Trend/
```

The repository contains a pre-built Vite/React application with only a `dist/` folder — no Node.js source. Static files are served directly via Nginx.

2.2 Run Application Locally (Port 3000)

```
docker run -d -p 3000:3000 trend-app:v1  
curl -I http://localhost:3000
```



3. Docker

3.1 Dockerfile

```
FROM nginx:alpine
COPY dist/ /usr/share/nginx/html
COPY nginx.conf /etc/nginx/conf.d/default.conf
EXPOSE 3000
CMD ["nginx", "-g", "daemon off;"]
```

3.2 nginx.conf

```
server {
    listen 3000;
    server_name _;
    root /usr/share/nginx/html;
    index index.html;
    location / {
        try_files $uri $uri/ /index.html;
    }
}
```

3.3 Build and Run

```
docker build -t trend-app:v1 .
docker run -d -p 3000:3000 trend-app:v1
docker ps
```

```
root@ip-172-31-11-132 trend# docker build -t trend-app:v1 .
[+] Building 4.0s (8/8) FINISHED
[+] [internal] load build definition from Dockerfile
[+] => transferring dockerfile: 24kB
[+] [internal] load metadata for docker.io/library/nginx:alpine
[+] [internal] load .dockerignore
[+] => transferring context: 2B
[+] [1/5] FROM docker.io/library/nginx:alpine@sha256:b1e7274a03d8b2c95171cc554396fe23ab0c11c596e27db910edde621e559a
[+] => resolve docker.io/library/nginx:alpine@sha256:b1e7274a03d8b2c95171cc554396fe23ab0c11c596e27db910edde621e559a
[+] => sha256:b1e7274a03d8b2c95171cc554396fe23ab0c11c596e27db910edde621e559a 10.30kB / 10.30kB
[+] => sha256:31db504b34376493f4d4e95d8a8403bbffaf6297fede32d0f923ce2dd523a 2.50kB / 2.50kB
[+] => sha256:4a944b3b9a7e2c23fe6f422e489dc288f9e7ced8e644b30249aaa3f0440 12.41kB / 12.41kB
[+] => sha256:6a8ac117861a477b048b7f930145215e031c0f09763135a70a4a8adda57bb 3.94kB / 3.94kB
[+] => sha256:abaaef6d1626e429b3f1209aa369c0a49562cc06b3e075c006e06f699ba3562 1.89kB / 1.89kB
[+] => sha256:41f2344650a2f2f123c7e76a302d22ed62cc02d4a51edca72ee732ef01bfacd 42kB / 42kB
[+] => extracting sha256:6a0ac1617861a477b048b7f930145215e031c0f09763135a70a4a8adda57bb
[+] => sha256:de1b677d80020e7e5852f0c046e2c2032424e0dd6f06042a369ad372257 957B / 957B
[+] => extracting sha256:abaaef6d1626e429b3f1209aa369c0a49562cc06b3e075c006e06f699ba3562
[+] => sha256:94d553cf70cab44ec7e90caba51c4d87b3d413617dbcf2fca440d935c14b04 405B / 405B
[+] => extracting sha256:41f2344650a2f2f123c7e76a302d22ed62cc02d4a51edca72ee732ef01bfacd
[+] => extracting sha256:de1b677d80020e7e5852f0c046e2c2032424e0dd6f06042a369ad372257
[+] => extracting sha256:94d553cf70cab44ec7e90caba51c4d87b3d413617dbcf2fca440d935c14b04
[+] => sha256:a302ff4a4b25725673e2fea2b3b9e2094c0b577d324e74abbf14470b5f1cfd 1.40kB / 1.40kB
[+] => sha256:fbaed1f7fcb6e40d591ac601924f1f03232ceet2741fde9e950dce5550caf8 20.20kB / 20.20kB
[+] => sha256:e55d4bbbb9e1a1fa030d24723aeeb60bc655f4eb4d6215a49db77a7d104697 1.21kB / 1.21kB
[+] => extracting sha256:e84d4bbbb9e1a1fa030d24723aeeb60bc655f4eb4d6215a49db77a7d104697
[+] => extracting sha256:a302ff4a4b25725673e2fea2b3b9e2094c0b577d324e74abbf14470b5f1cfd
[+] => extracting sha256:fbaed1f7fcb6e40d591ac601924f1f03232ceet2741fde9e950dce5550caf8
[+] [internal] load build context
[+] => transferring context: 9.24kB
[+] [2/3] COPY dist/ /usr/share/nginx/html
[+] [3/3] COPY nginx.conf /etc/nginx/conf.d/default.conf
[+] exporting to image
[+] => exporting layers
[+] => waiting image sha256:923d8c93e10db73a30dce67a1a72ef1d6a972260c7b3b5e6d05c23e1a0b49e4
[+] => naming to docker.io/library/trend-app:v1
root@ip-172-31-11-132 trend# docker run -d -p 3000:3000 trend-app:v1
root@ip-172-31-11-132 trend# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
fe20d5ef9aea	trend-app:v1	"/docker-entrypoint..."	5 hours ago	Up 5 hours	80/tcp, 0.0.0.0:3000->3000/tcp, :::3000->3000/tcp	vibrant_swanson

```
root@ip-172-31-11-132 trend#
```

4. DockerHub

```
docker tag trend-app:v1 sudirpandian01/trend-app:v1
```

```
docker push sudirpandian01/trend-app:v1
```

DockerHub Repository: <https://hub.docker.com/r/sudirpandian01/trend-app>

[Explore](#) / [sudirpandian01](#) / trend-app



sudirpandian01/trend-app
By [sudirpandian01](#) · Updated 13 minutes ago
[image](#)
☆ 0 ↓ 67

Overview **Tags**

Sort by **Newest**

TAG		
latest		<code>docker pull sudirpandian01/trend-app:latest</code>
Last pushed 13 minutes by sudirpandian01		
Digest	OS/ARCH	Compressed size
f1a7a8f5093f	linux/amd64	33.41 MB

TAG		
v5		<code>docker pull sudirpandian01/trend-app:v5</code>
Last pushed 14 minutes by sudirpandian01		
Digest	OS/ARCH	Compressed size
f1a7a8f5093f	linux/amd64	33.41 MB

5. Terraform Infrastructure

Infrastructure defined in `terraform/main.tf` provisions the Jenkins server in the default AWS VPC.

Resource	Type	Details
VPC	Data source	vpc-08cd654ebd17167be (default)
Security Group	aws_security_group	Ports 22, 8080, 3000 open
IAM Profile	Data source	Existing profile: sudir
EC2 Instance	aws_instance	t3.medium, Ubuntu, Jenkins pre-installed

```
terraform init
terraform validate
terraform plan
terraform apply -auto-approve
```

The screenshot shows the AWS IAM console interface. The top navigation bar includes the AWS logo, account name 'InPrivate', and a 'Terminate session' button. Below the navigation bar, there's a breadcrumb trail: 'Instances | EC2'. The main content area displays the 'Shortcuts' tab for the instance 'i-0035e0985144e541d (infra)'. The central part of the screen shows a JSON policy document for the user 'sudripandian01/'. At the bottom, a red box highlights the 'CheckResults' field, which contains Terraform output logs.

```
{
    "to_port": 2000
},
{
    "cidr_blocks": [
        "0.0.0.0/0"
    ],
    "description": "",
    "from_port": 8080,
    "ip_protocol": "tcp",
    "prefix_list_ids": [],
    "protocol": "tcp",
    "security_groups": [],
    "self": false,
    "to_port": 8080
}
],
"name": "trend-jenkins-sg",
"name_prefix": "",
"owner_id": "645437262580",
"region": "ap-south-1",
"revoke_rules_on_delete": false,
"tags": {
    "Name": "trend-jenkins-sg"
},
"tags_all": {
    "Name": "trend-jenkins-sg"
},
"timeouts": null,
"vpc_id": "vpc-08cd654ebd17167be"
},
"sensitive_attributes": [],
"identity_schema_version": 0,
"identity": {
    "account_id": "645437262580",
    "id": "sg-0dde492b3832c3e1b",
    "region": "ap-south-1",
    "private": "eyJhbGwYJmVjcmVlIyZPdHlwZTtYOGY4OC0xNDMGMzJjNmQWYkhaOmsiY3JlYXkiY3Rlc2MDAwMDAwMDAwImRlbGVGU2STSGOTBwMDAwMDAwfSwic2N0ZW1kaXZ2cllnbmh24iOiIxIn0=",
    "dependencies": [
        "data.aws.vpc.default"
    ]
}
},
},
},
"check_results": null
[{"out@sp-172-31-11-132 terraform$ terraform output
jenkins_public_ip = "15-233.6.179"
[{"out@sp-172-31-11-132 terraform$ }
```


6. Version Control

6.1 .gitignore

```
node_modules
.env
terraform.tfstate
terraform.tfstate.backup
.terraform/
*.pem
```

6.2 .dockerignore

```
node_modules
.git
.gitignore
README.md
*.pem
```

6.3 Push to GitHub

```
git init
git add .
git commit -m "Initial commit"
git branch -M project-2
git remote add origin https://github.com/sudirpandian/Devops.git
git push -u origin project-2 --force
```

The screenshot shows a GitHub repository interface for a project named 'project-2'. The repository is currently on the 'project-2' branch, which is 9 commits ahead of and 1 commit behind the 'main' branch. The repository has 13 branches and 0 tags. The file list shows the following files and their commit history:

File	Commit Message	Time Ago
root	Add README with setup instructions	da66ed6 - 19 minutes ago
dist	Project Files	4 months ago
k8s	Add k8s manifests and Jenkinsfile	3 hours ago
terraform	Fix Terraform user_data - Java 21 and correct Jenkins GPG ke...	3 hours ago
.dockerignore	Initial commit - trend-app with Dockerfile and nginx config	5 hours ago
.gitignore	Initial commit - trend-app with Dockerfile and nginx config	5 hours ago
Dockerfile	Initial commit - trend-app with Dockerfile and nginx config	5 hours ago
Jenkinsfile	Add KUBECONFIG to Jenkinsfile	2 hours ago
README.md	Add README with setup instructions	19 minutes ago
nginx.conf	Initial commit - trend-app with Dockerfile and nginx config	5 hours ago

The README file is titled 'Trend App - Production Deployment' and has a section for 'Architecture'.

On the right side of the repository page, there are sections for 'Releases' (No releases published), 'Packages' (No packages published), and 'Contributors' (1 contributor: sudirpandian).

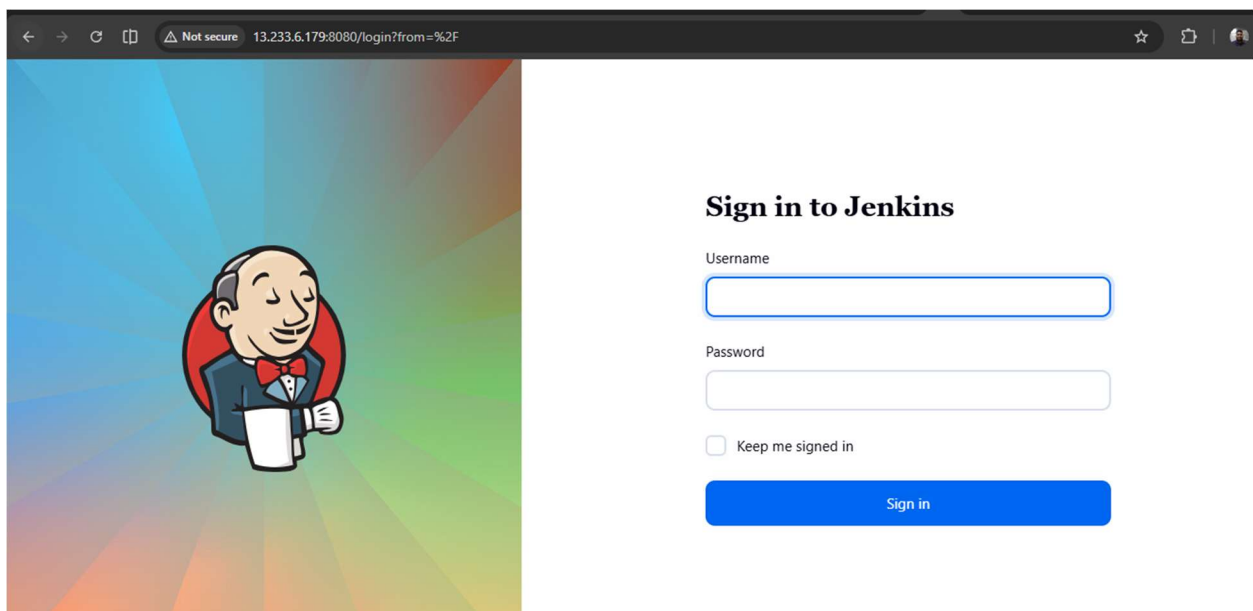
7. Jenkins

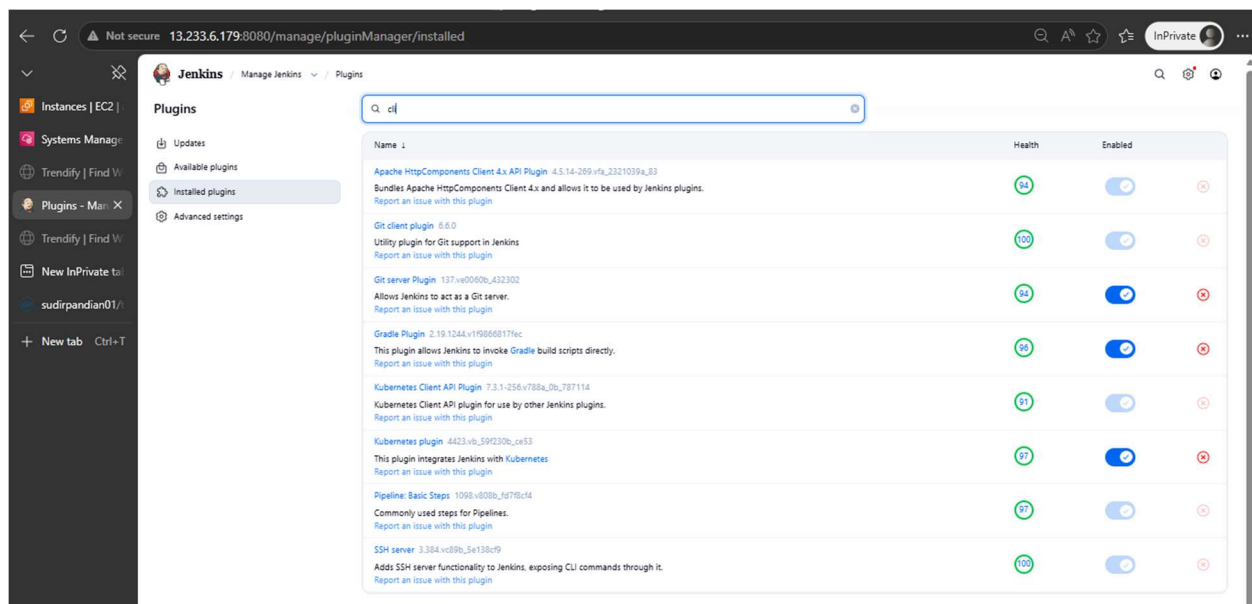
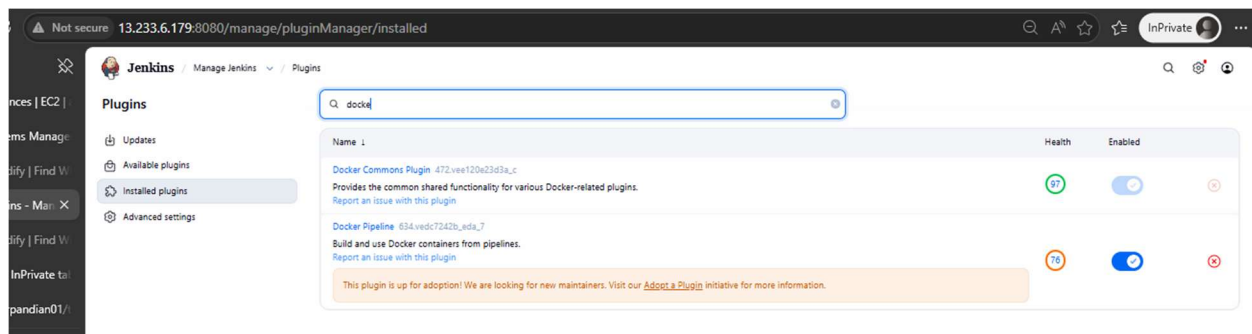
7.1 Installation

Jenkins is provisioned automatically via Terraform user_data on the EC2 instance at **13.233.6.179:8080**. Java 21 (OpenJDK) is required for Jenkins 2.555+.

7.2 Plugins Installed

Plugin	Purpose
Git	Source code checkout
Pipeline	Declarative pipeline support
Docker Pipeline	Docker build and push in pipeline
Kubernetes	Kubernetes integration
Kubernetes CLI	kubectl commands in pipeline
GitHub	GitHub integration
Credentials Binding	Secure credential injection





7.3 GitHub Webhook

Webhook configured at GitHub → Settings → Webhooks:

Payload URL: `http://13.233.6.179:8080/github-webhook/`

Content type: `application/json`

Trigger: Push events

8. CI/CD Pipeline

8.1 Pipeline Explanation

Stage	Action	Details
Checkout	Pull source code	Clones project-2 branch from GitHub
Docker Build	Build image	Tags with build number (v1, v2, v3...)
Docker Push	Push to DockerHub	Pushes versioned tag + latest
Deploy to EKS	Rolling update	kubectrl set image + rollout status

8.2 Jenkinsfile

```
pipeline {
  agent any
  environment {
    DOCKERHUB_USER = 'sudirpandian01'
    IMAGE_NAME = 'trend-app'
    IMAGE_TAG = "v${BUILD_NUMBER}"
    EKS_CLUSTER = 'trend-eks'
    AWS_REGION = 'ap-south-1'
    KUBECONFIG = '/var/lib/jenkins/.kube/config'
  }
  stages {
    stage('Checkout') { ... }
    stage('Docker Build') { ... }
    stage('Docker Push') { ... }
    stage('Deploy to EKS') { ... }
  }
}
```

Not secure 13.233.6.179:8080/job/trend-pipeline/5/stages/?selected-node=45

Jenkins / trend-pipeline / #5 / Stages

< #5

Started by GitHub push by sudirpandian Started 25 min ago Queued 6.4 sec Took 28 sec - Changes

Start → Checkout SCM → Checkout → Docker Build → Docker Push → Deploy to EKS → Post Actions → End

71ms Started 25m ago Jenkins

Deployment successful

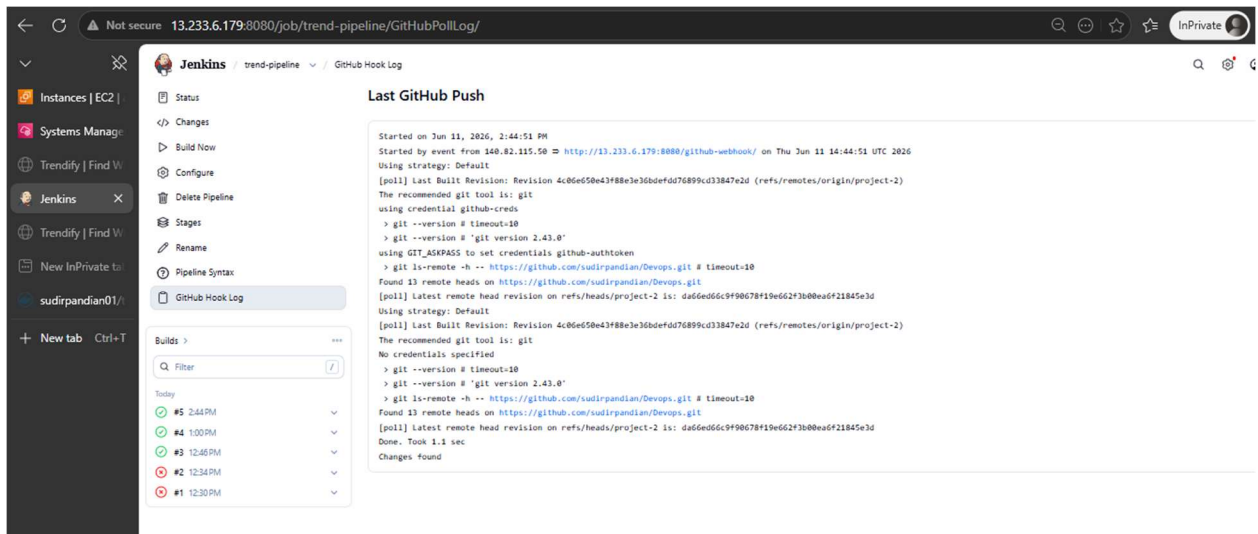
Deployment successful

Checkout SCM 1s
Checkout 0.89s
Docker Build 2s
Docker Push 11s
Deploy to EKS 10s
Post Actions 71ms

Not secure 13.233.6.179:8080/job/trend-pipeline/5/console

Jenkins / trend-pipeline / #5

```
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to EKS)
[Pipeline] sh
+ aws eks update-kubeconfig --region ap-south-1 --name trend-eks
Updated context arn:aws:eks:ap-south-1:645437362580:cluster/trend-eks in /var/lib/jenkins/.kube/config
[Pipeline] sh
+ kubectl set image deployment/trend-app trend-app-sudirpandian01/trend-app:v5
deployment.apps/trend-app image updated
[Pipeline] sh
+ kubectl rollout status deployment/trend-app
Waiting for deployment "trend-app" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "trend-app" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "trend-app" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "trend-app" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "trend-app" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "trend-app" rollout to finish: 1 old replicas are pending termination...
deployment "trend-app" successfully rolled out
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Deployment successful!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```



9. Kubernetes — AWS EKS

9.1 Cluster Creation

```
eksctl create cluster \
  --name trend-eks \
  --region ap-south-1 \
  --nodegroup-name workers \
  --node-type t3.medium \
  --nodes 2 \
  --managed
```

```
root@ip-172-31-11-132 trend]$ eksctl create cluster \
  --name trend-eks \
  --region ap-south-1 \
  --nodegroup-name workers \
  --node-type t3.medium \
  --nodes 2 \
  --managed
2026-06-11 11:38:03 [i] eksctl version 0.227.0
2026-06-11 11:38:03 [i] using region ap-south-1
2026-06-11 11:38:03 [i] setting availability zones to [ap-south-1c ap-south-1a ap-south-1b]
2026-06-11 11:38:03 [i] subnets for ap-south-1c - public:192.168.0.0/19 private:192.168.96.0/19
2026-06-11 11:38:03 [i] subnets for ap-south-1a - public:192.168.32.0/19 private:192.168.128.0/19
2026-06-11 11:38:03 [i] subnets for ap-south-1b - public:192.168.64.0/19 private:192.168.160.0/19
2026-06-11 11:38:03 [i] nodegroup "workers" will use "" (AmazonLinux2022/1.34)
2026-06-11 11:38:03 [i] Auto Mode will be enabled by default in an upcoming release of eksctl. This means managed node groups and managed networking add-ons will no longer
maintain current behavior. explicitly set 'autoModeConfig.enabled: false' in your cluster configuration. Learn more: https://eksctl.io/usage/auto-mode/
2026-06-11 11:38:03 [i] using Kubernetes version 1.34
2026-06-11 11:38:03 [i] creating EKS cluster "trend-eks" in "ap-south-1" region with managed nodes
2026-06-11 11:38:03 [i] will create 3 separate CloudFormation stacks for cluster itself and the initial managed nodegroup
2026-06-11 11:38:03 [i] if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=ap-south-1 --cluster=trend-eks'
2026-06-11 11:38:03 [i] Kubernetes API endpoint access will use default of (publicAccess=true, privateAccess=false) for cluster "trend-eks" in "ap-south-1"
2026-06-11 11:38:03 [i] CloudWatch logging will not be enabled for cluster "trend-eks" in "ap-south-1"
2026-06-11 11:38:03 [i] you can enable it with eksctl utils updateclusterlogging --enable-awslogs --set-type=awslogs --region=ap-south-1 --cluster=trend-eks
```

9.2 Verify Nodes

```
kubectl get nodes
```

```
root@ip-172-31-11-132 terraform]$ kubectl get nodes
NAME                                STATUS    ROLES    AGE     VERSION
ip-192-168-61-106.ap-south-1.compute.internal Ready    <none>   3h26m   v1.34.8-eks-3385e9b
ip-192-168-71-102.ap-south-1.compute.internal Ready    <none>   3h26m   v1.34.8-eks-3385e9b
root@ip-172-31-11-132 terraform]$
```

9.3 Kubernetes Manifests

Files: k8s/deployment.yaml and k8s/service.yaml

```
kubectl apply -f k8s/
kubectl get pods
kubectl get svc
```

```
[root@ip-172-31-11-132 terraform]# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
trend-app-7b56c84f8-4l8lq          1/1     Running   0           29m
trend-app-7b56c84f8-t8tkw          1/1     Running   0           29m
[root@ip-172-31-11-132 terraform]#
```

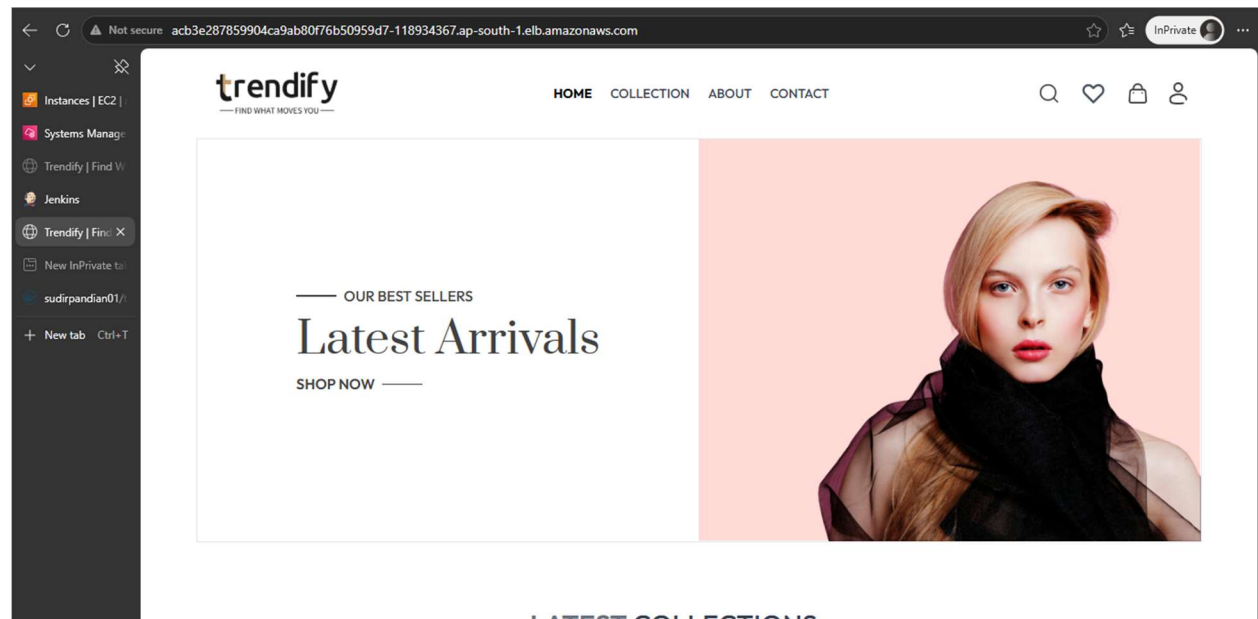
```
NAME          TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
kubernetes    ClusterIP     10.100.0.1      <none>           443/TCP          3h32m
trend-service LoadBalancer 10.100.164.171  acb3e287859904ca9ab80f76b50959d7-118934367.ap-south-1.elb.amazonaws.com 80:32485/TCP 3h19m
[root@ip-172-31-11-132 terraform]#
```

10. Application LoadBalancer

LoadBalancer DNS:

acb3e287859904ca9ab80f76b50959d7-118934367.ap-south-1.elb.amazonaws.com

Access URL: <http://acb3e287859904ca9ab80f76b50959d7-118934367.ap-south-1.elb.amazonaws.com>

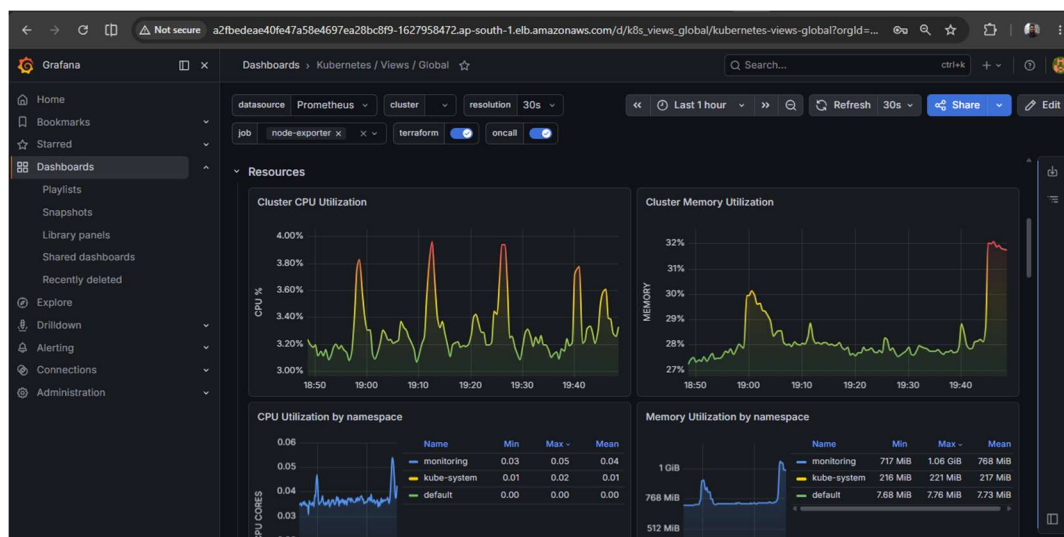
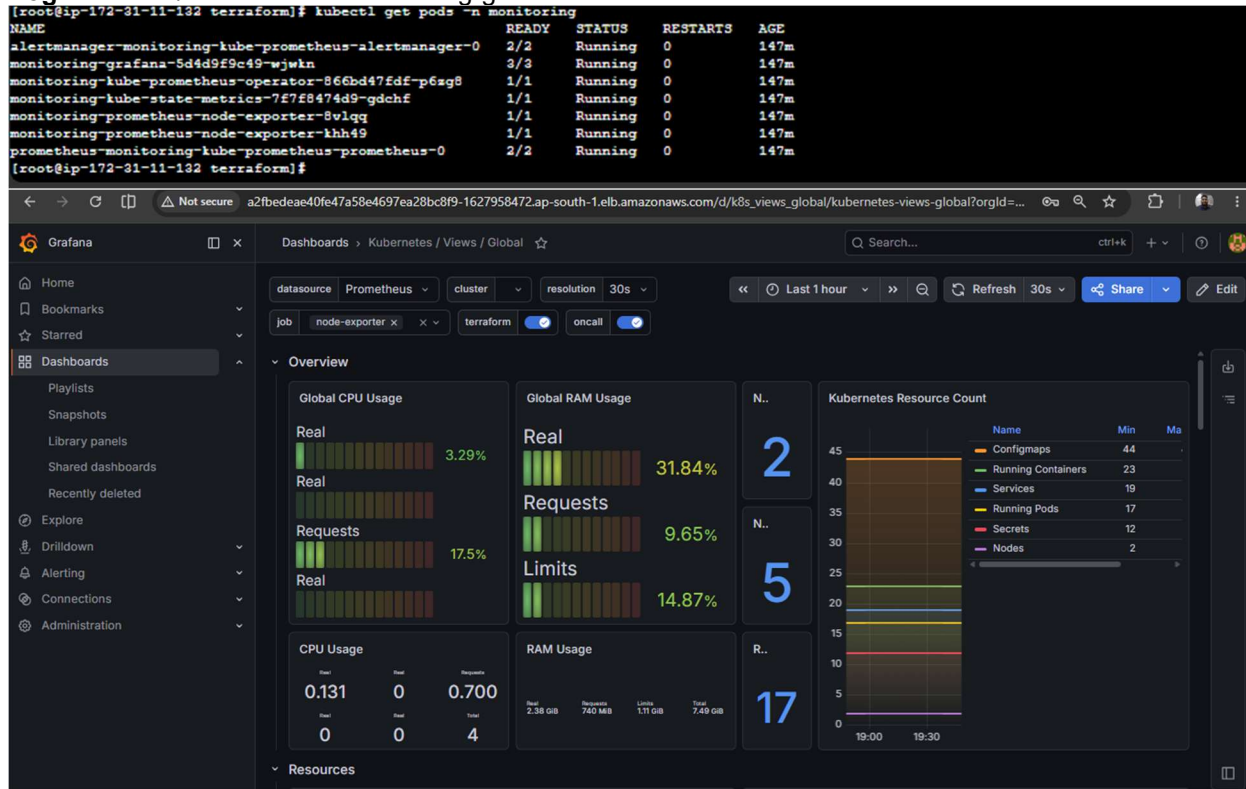


11. Monitoring

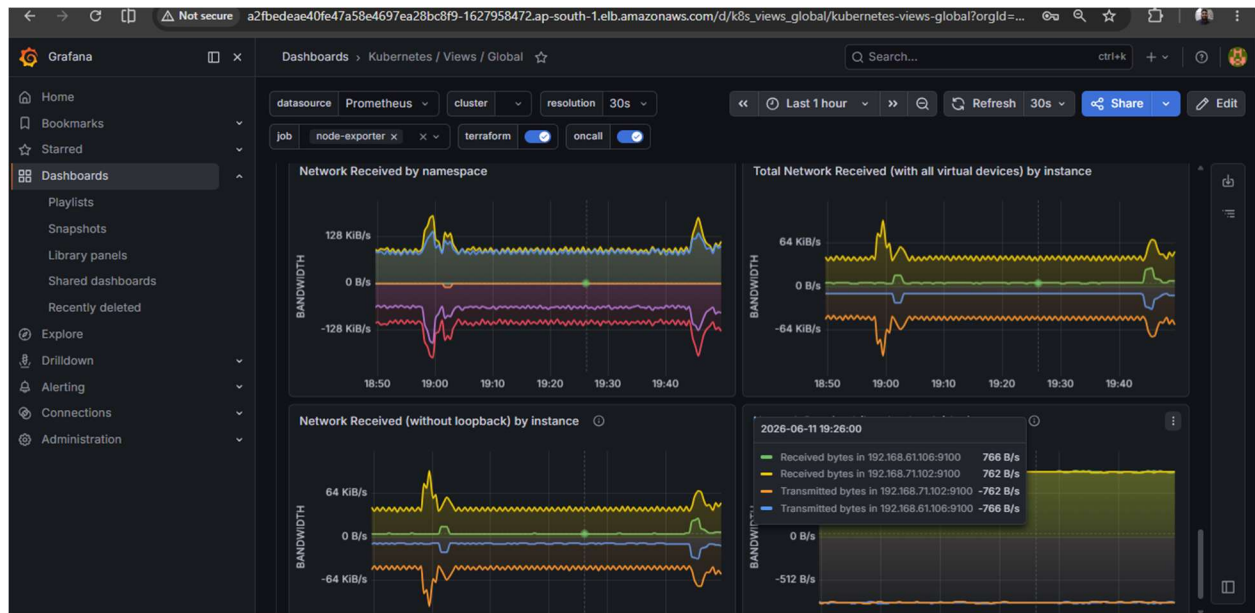
```
helm repo add prometheus-community https://prometheus-community.github.io/helm-charts
helm install monitoring prometheus-community/kube-prometheus-stack \
  --namespace monitoring --create-namespace
kubectl get pods -n monitoring
```

Grafana URL: <http://a2fbedeae40fe47a58e4697ea28bc8f9-1627958472.ap-south-1.elb.amazonaws.com>

Login: admin / <secret from monitoring-grafana secret>







12. Submission Summary

GitHub Repository	https://github.com/sudirpandian/Devops/tree/project-2
DockerHub Image	sudirpandian01/trend-app
Jenkins Server	http://13.233.6.179:8080
EKS Cluster	trend-eks (ap-south-1)
App LoadBalancer ARN	acb3e287859904ca9ab80f76b50959d7-118934367.ap-south-1.elb.amazonaws.com
Grafana	a2fbdeae40fe47a58e4697ea28bc8f9-1627958472.ap-south-1.elb.amazonaws.com